Physics 468/668: Statistical Physics **Spring 2019**

Lecture 6: The Microcanonical ensemble

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6.1 Properties

In this ensemble, the macrostate of a system is defined by (N, V, E). However, in view of the considerations in the previous lecture, we may prefer to specify a range of energy values, say from $(E - 1/2\Delta)$ to $(E + 1/2\Delta)$. With the macrostate specified, a choice still remains for a system to be in any one of a large number of possible microstates. In the phase space, the representative points of the ensemble have a choice to lie anywhere within a 'hypershell' defined by the conditions.

$$E - \frac{1}{2}\Delta \leq H(q,p) \leq E + \frac{1}{2}\Delta \tag{6.1}$$

The volume of the phase space enclosed within this shell is given by

$$\omega = \int d\omega = \int (d^{3N}q d^{3N}p) \tag{6.2}$$

It is clear that ω would be a function of (N, V, E, Δ).

Now, we consider the density function,

$$\begin{aligned} \rho(q,p) &= \text{const} \quad \text{if} \quad (E - 1/2\Delta) \leq H(q,p) \leq (E + 1/2\Delta) \\ &= 0 \qquad \text{otherwise} \end{aligned} \tag{6.3}$$

Accordingly, we expect that the number of representative points lying in a hypershell with a thickness of Δ.

Since the ensemble under study is a stationary one, the ensemble average of any property f will be independent of time. The next thing is to establish a connection between the mechanics and thermodynamics of the systems. The volume ω is a direct measurement of the multiplicity (Γ) of the microstates accessible to the system.

Let's say,

$$\Gamma = \omega/\omega_0 \tag{6.4}$$

The thermodynamics of the system would then follow in the same way as

$$S = k\ln\Gamma = k\ln(\omega/\omega_0) \tag{6.5}$$

The problem then is to determine ω_0. From the dimensional considerations, ω_0 must be in the nature of an *angular momentum raised to the power 3N*. This could be determined exactly for a few simple systems.

6.2 Ideal gas

For ideal gas in the micro-canonical ensemble, the volume ω of the phase space is given by

$$\omega = \int \cdots \int (d^{3N}q d^{3N}p) \tag{6.6}$$

Since the Hamiltonian in this case is a function of the p_i alone, integrations over q_i could be carried out as follows,

$$\omega = \int \cdots \int_{E-\frac{1}{2}\Delta \le \sum_{i=1}^{3N}(p_i^2/2m) \le E+\frac{1}{2}\Delta} d^{3N}p = \int \cdots \int_{2m(E-\frac{1}{2})\Delta \le \sum_{i=1}^{3N}(p_i^2) \le 2m(E+\frac{1}{2}\Delta)} d^{3N}y \tag{6.7}$$

This is equal to the volume of $3N$-dimensional hyper-shell bounded by a hypersphere of radii

$$\sqrt{2m(E-\frac{1}{2})\Delta} \le r \le \sqrt{2m(E+\frac{1}{2}\Delta)} \tag{6.8}$$

If $\Delta \ll E$, this is given by the thickness of the shell, which is almost equal to $\Delta(m/2E)^{1/2}$, multiplied by the surface area of a $3N$-dimensional hyper-shphere. By the equation of (7) of Appendix C, we obtain the integral,

$$\Delta\left(\frac{m}{2E}\right)^{1/2}\left[\frac{2\pi^{3N/2}}{[(3N/2-1)]!}(2mE)^{(3N-1)/2}\right] \tag{6.9}$$

Therefore,

$$\omega \approx \frac{\Delta}{E}V^N \frac{(2\pi mE)^{3N/2}}{[(3N/2-1)!]} \tag{6.10}$$

Comparing this with the following

$$\omega_0 = h^{3N} \tag{6.11}$$

Where the upper-script denotes the number of degrees of freedom.

In the case of a single particle (N=3), the number of microstates available would asymptotically be equal to the volume of allowed region divided by h^3. We then calculate the number of microstates available to a free particle confined to volume V and its momentum p being less than or equal to a given value P,

$$\sum(P) \approx \frac{1}{h^3}\int \cdots \int_{p \le P}(d^3q d^3p) = \frac{V}{h^3}\frac{4\pi}{3}P^3, \tag{6.12}$$

from which we obtain for the number of microstates with momentum lying between p and $p+dp$,

$$g(p)dp = \frac{d\sum(p)}{dp}dp \approx \frac{V}{h^3}4\pi p^2 dp \tag{6.13}$$

Expressed in terms of the particle energy, these expressions assume the form

$$\sum(E) \approx \frac{V}{h^3}\frac{4\pi}{3}(2mE)^{3/2} \tag{6.14}$$

and

$$g(E)dE = \frac{d\sum(E)}{dE}dE \approx \frac{V}{h^3}2\pi(2m)^{3/2}E^{1/2}dE \tag{6.15}$$

6.3 1D simple harmonic oscillator

For this system, we have the Hamiltonian as follows

$$H(q,p) = \frac{1}{2}kq^2 + \frac{1}{2m}p^2 \tag{6.16}$$

where k is the spring constants. The space coordinate q and the momentum coordinate p of the system are given by

$$q = A\cos(\omega t + \phi) \tag{6.17}$$

$$p = m\dot{q} = -m\omega A\sin(\omega t + \phi) \tag{6.18}$$

Where A is the amplitude and ω is the angular frequency of vibration;

$$\omega = \sqrt{k/m} \tag{6.19}$$

And the energy of the oscillator is a constant of the motion,

$$E = \frac{1}{2}m\omega^2 A^2 \tag{6.20}$$

The phase-space trajectory of the representative point (q,p) of the system is determined by eliminating t ,

$$\frac{q^2}{(2E/m\omega^2)} + \frac{p^2}{2mE} = 1 \tag{6.21}$$

which is an ellipse, with axes proportional to $\sqrt{E}$. Now, if we restrict the oscillator energy to the interval $[E - 1/2\Delta, E + 1/2\Delta]$, the representative points in the phase space will be confined to the region bounded by elliptical trajectories corresponding to the energy values $E - 1/2\Delta$ and $E + 1/2\Delta$. The volume will be

$$\int\cdots\int_{E-\frac{1}{2}\Delta\le H(q,p)\le E+\frac{1}{2}\Delta} dqdp = \frac{2\pi(E+\frac{1}{2}\Delta)}{\omega} - \frac{2\pi(E-\frac{1}{2}\Delta)}{\omega} = \frac{2\pi\Delta}{\omega} \tag{6.22}$$

According to quantum mechanics, the energy eigenvalues are,

$$E_n = (n + \frac{1}{2})\bar{h}\omega \tag{6.23}$$

In terms of phase phase, one could say that the representative point of the system must move along one of the 'chosen' trajectories; the area of the phase space between two consecutive trajectories, for which $\Delta = \bar{h}w$ is simply $2\pi\hbar$. For $E \gg \Delta \gg \hbar\omega$, the number of eigenstates within the allowed energy interval is nearly equal to $\Delta/\hbar\omega$. Hence, the area of the phase space equivalent to one eigenstate is given by,

$$\omega_0 = (2\pi\Delta/\omega)/(\Delta/\bar{h}w) = 2\pi\bar{h} = h \tag{6.24}$$

If we consider a system of N harmonic oscillators along the same lines, we arrive at the results: $\omega_0 = h^N$.

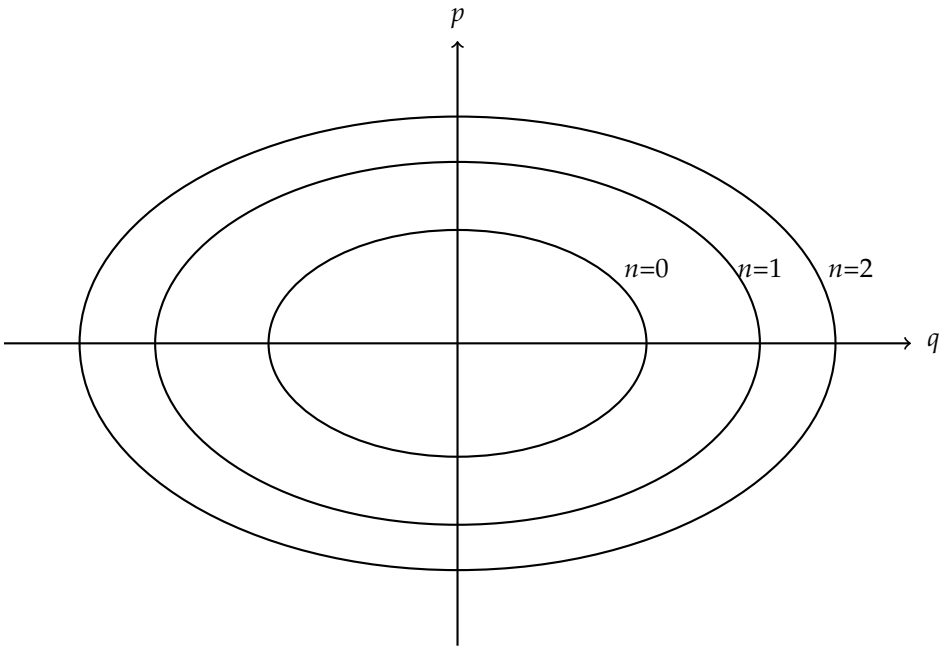


Figure 6.1: A schematic of phase space.

6.4 Homework

1. 2.3
2. 2.5
3. 2.7
4. 2.8
5. 2.9